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PROJET SAÉ : LOT C

Découverte de machine

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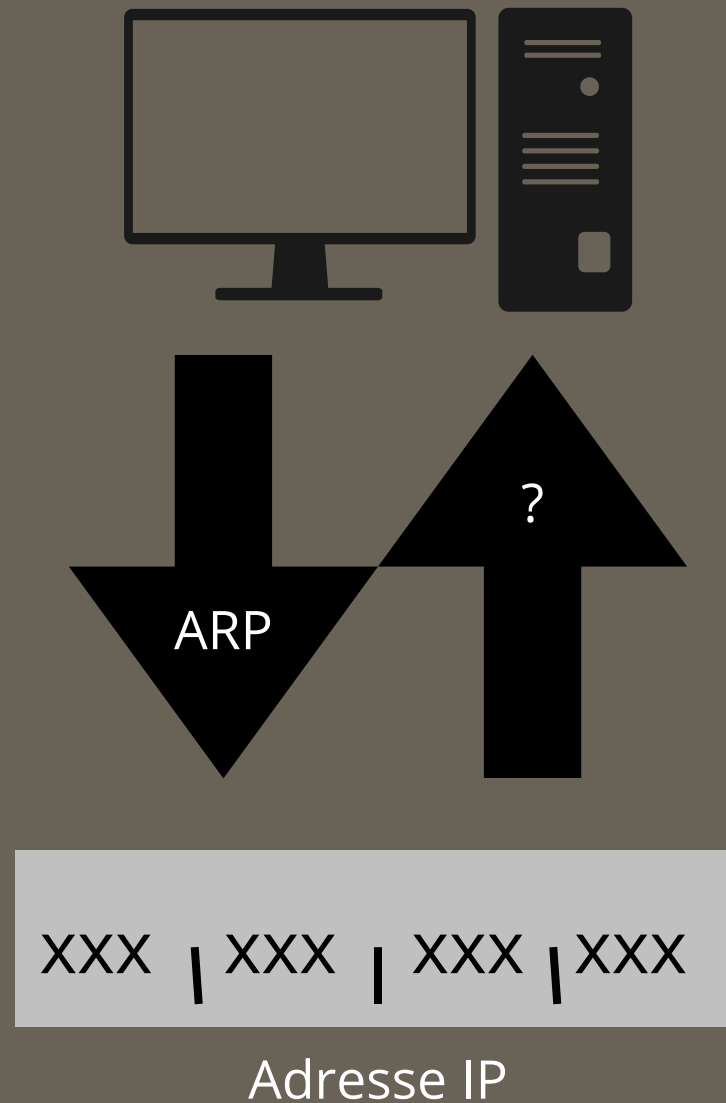
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Découvrir les machines du réseau

Utilisation de Arp-Scan



```
adminetu@RTP35:~$ sudo arp-scan --localnet
Interface: eno1, type: EN10MB, MAC: 84:69:93:07:a3:51, IPv4: 172.21.1.35
Starting arp-scan 1.10.0 with 65536 hosts (https://github.com/royhills/arp-scan)
172.21.0.1      e4:f0:04:ff:fa:9e      Dell Inc.
172.21.1.21     84:69:93:0a:01:c4      HP Inc.
172.21.1.22     84:69:93:09:ff:dc      HP Inc.
172.21.1.23     84:69:93:06:51:f1      HP Inc.
172.21.1.26     84:69:93:0a:00:91      HP Inc.
172.21.1.27     84:69:93:0a:01:ba      HP Inc.
172.21.1.28     84:69:93:0a:01:39      HP Inc.
172.21.1.30     84:69:93:0a:01:03      HP Inc.
172.21.1.33     84:69:93:0a:01:ac      HP Inc.
172.21.1.34     84:69:93:07:a3:6b      HP Inc.
172.21.1.60     24:6a:0e:77:16:a3      (Unknown)
172.21.1.63     24:6a:0e:77:1a:73      (Unknown)
172.21.1.64     24:6a:0e:77:28:8c      (Unknown)
172.21.1.65     24:6a:0e:76:f8:d4      (Unknown)
172.21.1.66     24:6a:0e:77:15:9b      (Unknown)
172.21.1.67     24:6a:0e:76:f8:fc      (Unknown)
172.21.1.68     24:6a:0e:76:f6:bc      (Unknown)
172.21.1.71     24:6a:0e:76:f8:74      (Unknown)
172.21.1.73     24:6a:0e:76:f8:df      (Unknown)
172.21.1.74     24:6a:0e:77:16:f8      (Unknown)
172.21.1.72     24:6a:0e:76:f6:ef      (Unknown)
172.21.1.75     24:6a:0e:76:f9:02      (Unknown)
172.21.1.76     24:6a:0e:77:16:86      (Unknown)
172.21.1.112    24:6a:0e:76:f8:ec      (Unknown)
172.21.1.117    24:6a:0e:77:17:25      (Unknown)
172.21.1.121    24:6a:0e:76:f5:d9      (Unknown)
172.21.1.126    24:6a:0e:77:17:33      (Unknown)
172.21.1.127    24:6a:0e:77:17:2a      (Unknown)
```

Découvrir les OS des machines sur le réseau (fingerprint)

Scan d'OS avec Nmap

```
Fichier  Édition  Affichage  Terminal  Onglets  Aide
adminetu@RTP23:~$ sudo nmap -O 172.21.1.27
[sudo] Mot de passe de adminetu :
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-21 16:34 CEST
Nmap scan report for 172.21.1.27
Host is up (0.00036s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
902/tcp   open  iss-realsecure
MAC Address: 84:69:93:0A:01:BA (HP)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.6
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.64 seconds
adminetu@RTP23:~$
```

Scan d'OS avec Nmap (active ou agressif)

```
adminetu@RTP23:~$ sudo nmap -A 172.21.1.27
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-21 15:33 CEST
Nmap scan report for 172.21.1.27
Host is up (0.00039s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 9.2p1 Debian 2+deb12u7 (protocol 2.0)
| ssh-hostkey:
|   256 e70a47840b02cdb7afe4611e7858cbb8 (ECDSA)
|_  256 06ee9544eedaae7380a5059829dea6f4 (ED25519)
902/tcp   open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
MAC Address: 84:69:93:0A:01:BA (HP)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.6
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1   0.39 ms  172.21.1.27

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.48 seconds
```

scan passive

sudo p0f -i eno1

```
Fichier Edition Arrimage Terminal Onglets Aide
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
   link/ether 24:6a:0e:77:16:cd brd ff:ff:ff:ff:ff:ff
   altname enp0s31f6
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DEFAULT group default
   link/ether 26:4f:83:ca:0c:13 brd ff:ff:ff:ff:ff:ff
adminetu@RTN06:~$ sudo p0f -i eno1
--- p0f 3.09b by Michal Zalewski <lcamtuf@coredump.cx> ---

[+] Closed 1 file descriptor.
[+] Loaded 322 signatures from '/etc/p0f/p0f.fp'.
[+] Intercepting traffic on interface 'eno1'.
[+] Default packet filtering configured [+VLAN].
[+] Entered main event loop.

.-[ 172.21.1.116/46760 -> 3.233.158.114/443 (syn) ]-
|
| client   = 172.21.1.116/46760
| os       = Linux 2.2.x-3.x
| dist     = 0
| params   = generic
| raw_sig  = 4:64+0:0:1460:mss*44,7:mss,sok,ts,nop,ws:df,id+:0
|
| ----
.-[ 172.21.1.116/46760 -> 3.233.158.114/443 (mtu) ]-
|
| client   = 172.21.1.116/46760
| link     = Ethernet or modem
| raw_mtu  = 1500
|
| ----
.-[ 172.21.1.116/46760 -> 3.233.158.114/443 (syn+ack) ]-
|
| server   = 3.233.158.114/443
| os       = ???
| dist     = 25
| params   = tos:0x08
| raw_sig  = 4:39+25:0:1400:mss*23,9:mss,sok,ts,nop,ws:df:0
|
```

```
Fichier Edition Arrimage Terminal Onglets Aide
|
| ----
.-[ 172.21.1.116/46760 -> 3.233.158.114/443 (syn+ack) ]-
|
| server   = 3.233.158.114/443
| os       = ???
| dist     = 25
| params   = tos:0x08
| raw_sig  = 4:39+25:0:1400:mss*23,9:mss,sok,ts,nop,ws:df:0
|
| ----
.-[ 172.21.1.116/46760 -> 3.233.158.114/443 (mtu) ]-
|
| server   = 3.233.158.114/443
| link     = generic tunnel or VPN
| raw_mtu  = 1440
|
| ----
.-[ 172.21.1.116/46760 -> 3.233.158.114/443 (uptime) ]-
|
| client   = 172.21.1.116/46760
| uptime   = 20 days 9 hrs 1 min (modulo 49 days)
| raw_freq = 1000.00 Hz
|
| ----
^C[!] WARNING: User-initiated shutdown.

All done. Processed 546 packets.
adminetu@RTN06:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 24:6a:0e:77:16:cd brd ff:ff:ff:ff:ff:ff
```


Découvrir les services disponibles

Utilisation de nmap

commandes :

`sudo nmap -sS -p- IP`

`sudo nmap -sV IP`

Etats possibles des ports :

- ouvert
- fermé

Utilisation de nmap

```
adminetu@RTP30:~$ nmap -sV unicaen.fr
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-21 16:34 CEST
Nmap scan report for unicaen.fr (193.55.120.26)
Host is up (0.0012s latency).
rDNS record for 193.55.120.26: rp5.unicaen.fr
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http   nginx
113/tcp    closed ident
443/tcp    open  ssl/http nginx
444/tcp    closed snpp
843/tcp    closed unknown
8443/tcp   closed https-alt

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.61 seconds
```

Utilisation de netcat

```
nc -zv [adresse IP cible] [num port ou intervalle]
```

Cible : passerelle

```
user@debian-bullseye:~$ nc -zv 172.21.0.1 22
_gateway [172.21.0.1] 22 (ssh) open
user@debian-bullseye:~$ nc -zv 172.21.0.1 1-1024
_gateway [172.21.0.1] 860 (?) : Connection timed out
_gateway [172.21.0.1] 22 (ssh) open
```

Découvrir les versions des services disponible

- Nmap + -sV

```
adminetu@LinuxDebianGUI:~$ nmap -Pn -sV 172.17.239.221
Starting Nmap 7.93 ( https://nmap.org ) at 2024-10-18 12:22 CEST
Nmap scan report for 172.17.239.221
Host is up (0.00028s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE          VERSION
80/tcp    open  http             Apache httpd 2.4.62 ((Debian))
902/tcp   open  ssl/vmware-auth  VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
```

- -sV : envoie une série de probes variées, compare réponses à signature
→ identification automatique et souvent version.
- --script=banner : complète en récupérant bannières brutes (parfois sV n'obtient pas suffisamment d'infos).

- Nmap + -script=banner

```
adminetu@LinuxDebianGUI:~$ nmap -Pn --script=banner 172.17.239.221
Starting Nmap 7.93 ( https://nmap.org ) at 2024-10-18 12:34 CEST
Nmap scan report for 172.17.239.221
Host is up (0.00020s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE
80/tcp    open  http
902/tcp   open  iss-realsecure
| banner: 220 VMware Authentication Daemon Version 1.10: SSL Required, Se
|_rverDaemonProtocol:SOAP, MKSDisplayProtocol:VNC , VMXARGS supported,...
```

- -Pn permet de ne pas rater une machine qui désactive/ignore ICMP ou bloque les probes de découverte ; utile sur des réseaux protégés.
- -sV permet d'obtenir la version du service (quand possible) — indispensable pour savoir si un service est vulnérable (ex : Apache 2.4.41 connu vulnérable X).

- NetCat

```
C:\Users\Utilisateur>ncat 192.168.47.98 22  
SSH-2.0-OpenSSH_9.2p1 Debian-2+deb12u2
```

Netcat est un outil léger qui ouvre des connexions TCP/UDP pour récupérer des bannières, tester des ports et transférer des données — à utiliser uniquement sur des cibles autorisées.

Modifier la bannière :

« ServerTokens »

HTTP :
/etc/apache2/apache2.conf

On ajoute/modifie :
ServerTokens [option]

Les options :
Prod, Major, Minor, Minimal, OS

Pour limiter la fuite d'information sur les serveurs HTTP, Apache propose ServerTokens et ServerSignature. Avec ServerTokens Prod, la bannière ne révèle que le nom du serveur, ce qui réduit le risque que des scanners détectent la version exacte et exploitent des vulnérabilités connues.

Conclusion

sources

Référence 1 : manuel arp-scan

Référence 2 : <https://nmap.org/docs.html>

Référence 3: <https://nmap.org/book/man-version-detection.html>

Référence 4: <https://man7.org/linux/man-pages/man1/nc.1.html>

UIT DE CAEN CAMPUS III

Merci !